Weights → np.random.random (0.01)

↳ default ⇒ sequential model

Weights initialization

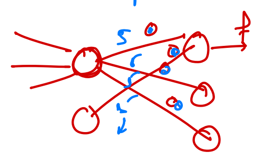
1. weight should be small
2. weights assigned should be different

no. Many weights should be different.

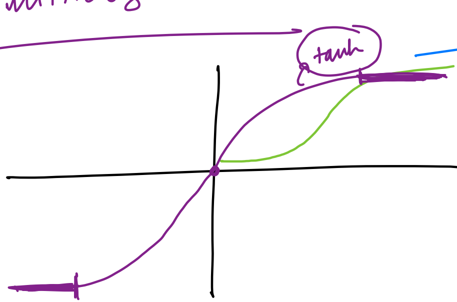
0.01 x 0.002 . . . . ⇒ vanish

exploding

3. ~~weights should be different~~



Weight initialization methods.

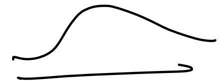


0.25

⇓

vanishing gradient

tanh   Xavier / glorot initialization



1. glorot normal → ND

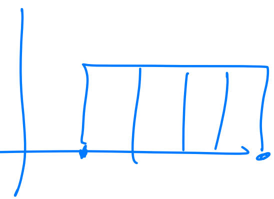
$$\sigma = \frac{2}{\text{fan in} + \text{fan out}}$$

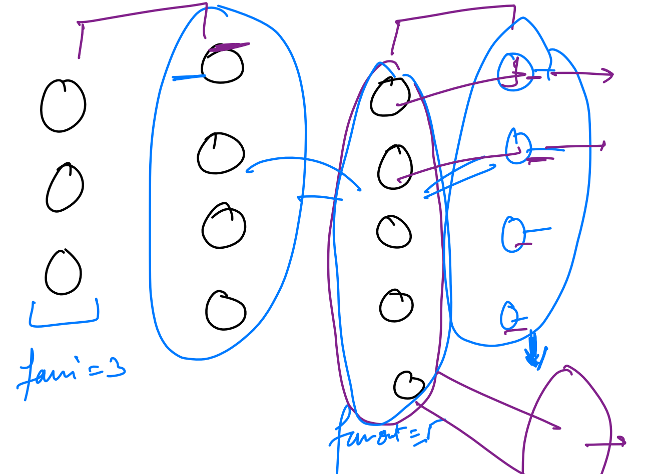
fan in → no. of neuron in previous

fan out → no. of neurons in next layer

2. glorot uniform

$$\left[\frac{-\sqrt{6}}{\sqrt{\text{fanin} + \text{fanout}}}, \frac{\sqrt{6}}{\sqrt{\text{fanin} + \text{fanout}}} \right]$$





Relu → He

He Normal → $N(0, \sigma)$ where $\sigma = \frac{2}{\text{fanin}}$

He Uniform → $\left[\frac{-\sqrt{6}}{\sqrt{\text{fanin}}}, \frac{\sqrt{6}}{\sqrt{\text{fanin}}} \right]$

input
↓
Layer 1
↓
Layer 2
↓
Layer 3
↓
output

skip connection

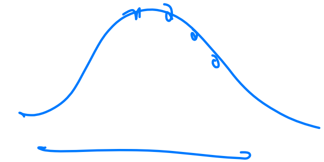
× sequential

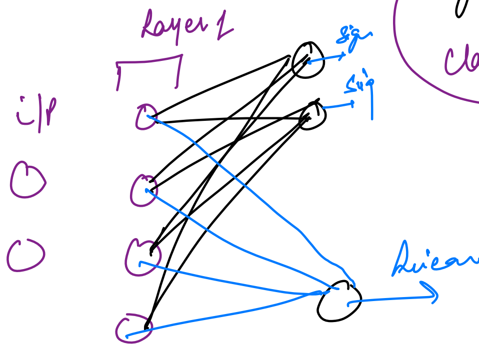
input
↓
Layer 1
↓
Layer 2
↓
Layer 3

Sigmoid
↓
Classification

Linear
↓
Regression

0.2





Functional API